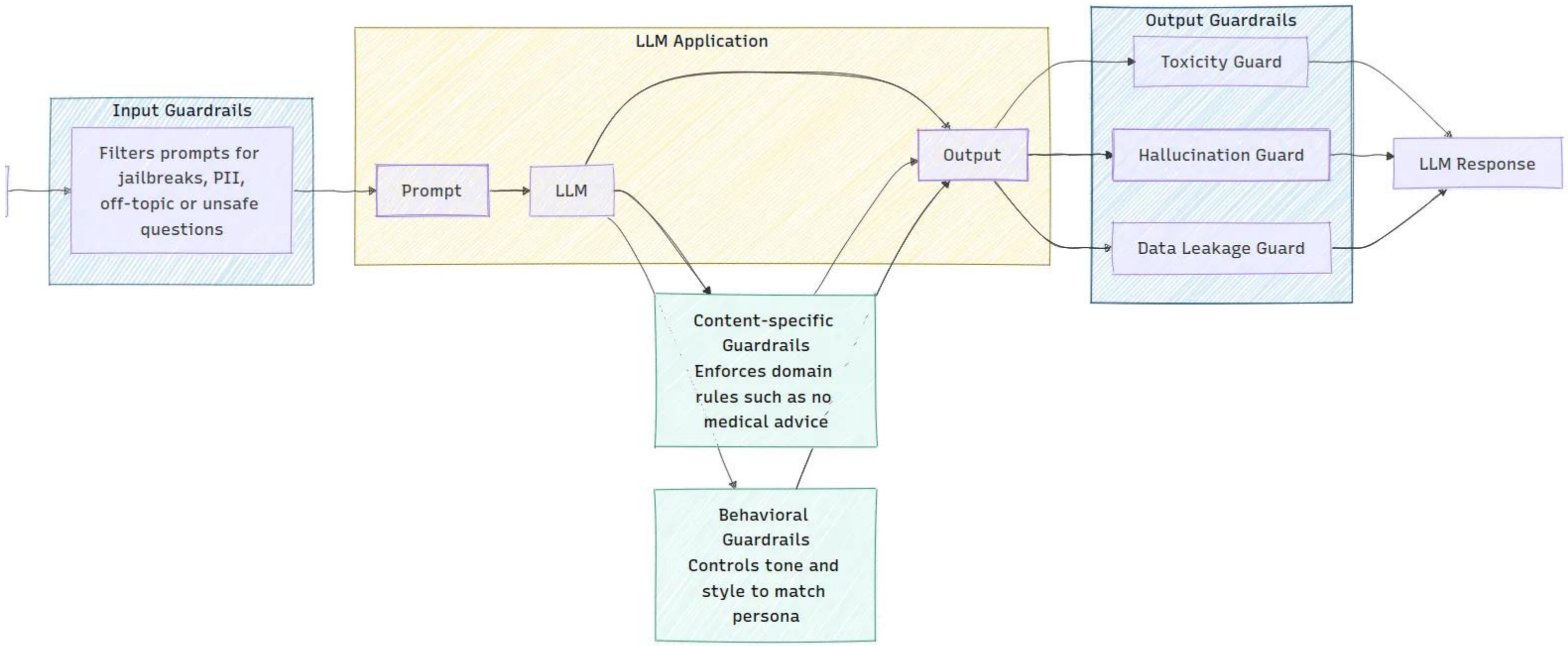
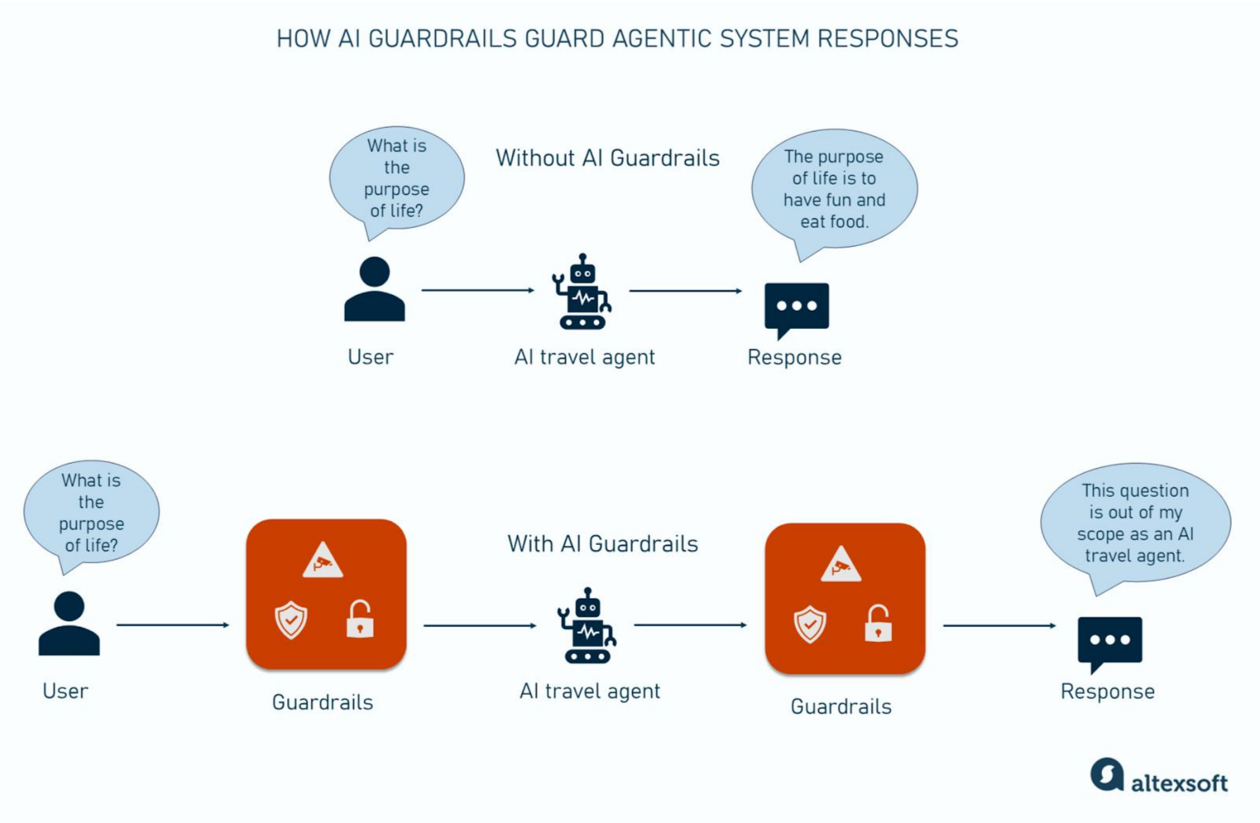
Module 7 – LLMOps & Production Guardrails



- **The Goal:** Moving from a "working prototype" to a "reliable utility."
- **The Missing Layer:** In production, AI systems need boundaries to prevent hallucinations, runaway costs, and safety violations.
- **Core Focus:** Guardrails, Inference Optimization, and Lifecycle Management (CI/CD).

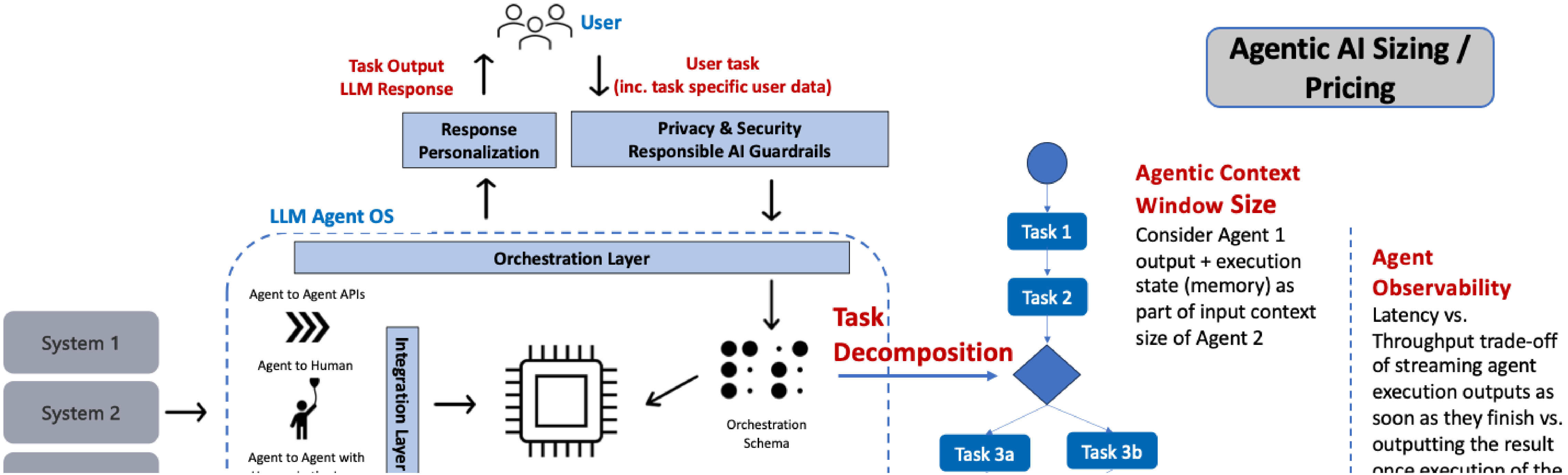
AI Guardrails & Safety



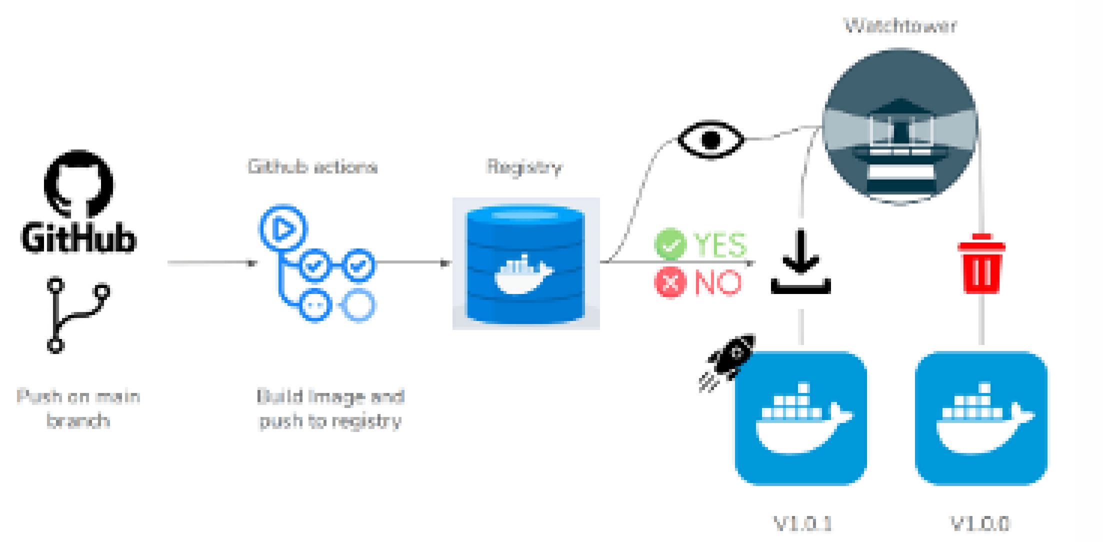
- **Boundary Enforcement:** Using **Guardrails AI** or **NeMo Guardrails** to validate inputs and outputs in real-time.
- **Topical Rails:** Ensuring the agent never discusses off-topic subjects (e.g., a Medical Agent shouldn't give investment advice).
- **Hallucination Filters:** Automatically rejecting outputs that don't match the "Memory Vault" context.

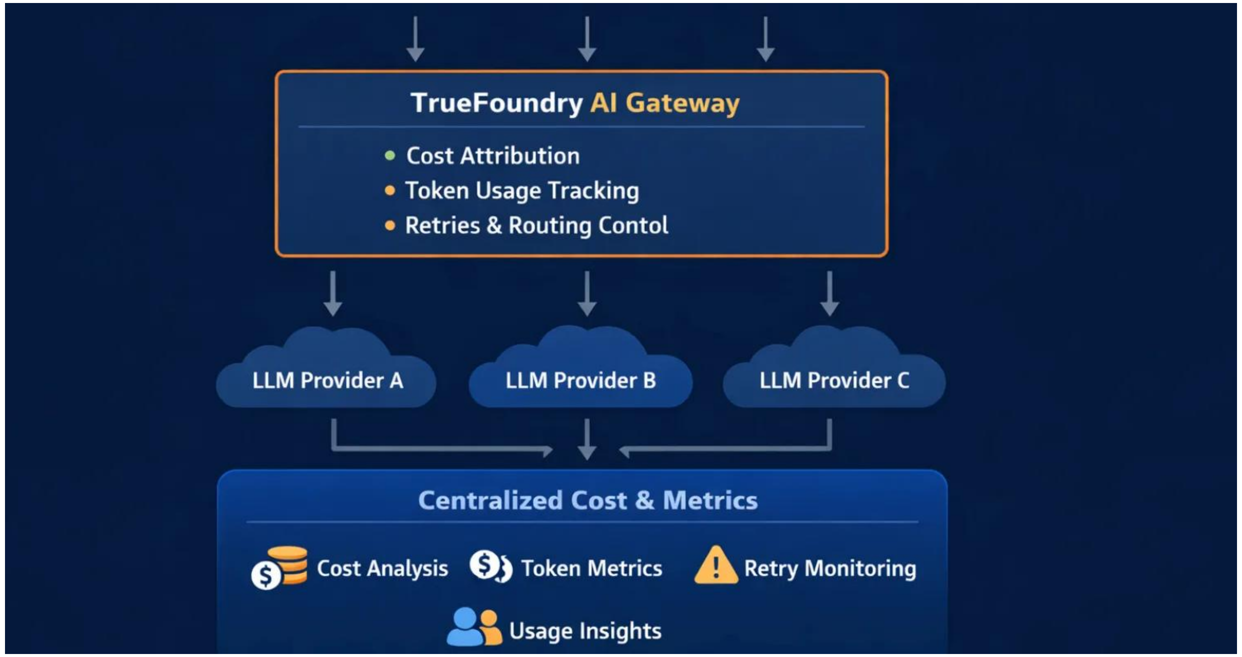
Inference Optimization at Scale

- **Throughput vs. Latency:** Utilizing **vLLM** and **TensorRT-LLM** for high-concurrency environments.
- **Quantization (GGUF/AWQ):** Compressing model weights to run Mistral-Large on smaller GPUs without losing technical precision.
- **Rate Limiting:** Protecting your API from being overwhelmed by too many simultaneous agent loops.



CI/CD & Industrial Observability





- **Automated Regression Testing:** Every time you update the prompt, a **GitHub Action** runs 50 test cases to ensure the agent's accuracy didn't drop.
- **Full-Trace Logging:** Capturing every "Thought" and "Action" into a central dashboard for post-failure analysis.
- **Cost Tracking:** Real-time monitoring of token usage to prevent unexpected billing spikes.